

MATH328 Reference

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1 Linear Algebra

1.1 Basics

The **zero matrix** is a matrix of all zeros. The **identity matrix** I is a square matrix of all ones on the main diagonal and zeros elsewhere.

The **rank** of a matrix is the number of non-zero rows in the echelon form of a matrix.

1.2 Matrix Multiplication

The product of two matrices A and B is defined as

$$AB = \left(\sum_{k=1}^n a_{ik} b_{kj} \right)$$

Note: the inner dimensions must match for matrix multiplication to be defined. Matrix multiplication is *not* commutative, i.e., $AB \neq BA$ in general. The **commutator** of two matrices A and B is defined as

$$[A, B] = AB - BA.$$

If $[A, B] = 0$, then A and B **commute**.

1.3 Linear Functions and Operators

A function or operator $T(\mathbf{x}) = A\mathbf{x}$ is linear iff

- $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ for any vectors $\mathbf{x}$ and $\mathbf{y}$, and
- $T(c\mathbf{x}) = cT(\mathbf{x})$ for any scalar c and vector $\mathbf{x}$.

1.4 Elementary Row Operations

- Swap two rows.
- Multiply a row by a nonzero scalar.
- Add a multiple of one row to another row.

These help solve matrix equations such as $A\mathbf{x} = \mathbf{b}$.

1.5 Consistency and Uniqueness of Solutions

Given a matrix equation $A\mathbf{x} = \mathbf{b}$, where n is the number of columns in A :

- $\text{rank}(A) < \text{rank}(A \mid \mathbf{b}) \implies$ **inconsistent**.
- $\text{rank}(A) = \text{rank}(A \mid \mathbf{b}) = n \implies$ **unique solution**.
- $\text{rank}(A) = \text{rank}(A \mid \mathbf{b}) = R < n \implies$ **infinitely many solutions**, with $n - R$ free variables.

1.6 Linear Independence

A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is **linearly independent** if the only solution to the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

is $c_1 = c_2 = \dots = c_n = 0$. In other words,

$$\text{rank} \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{pmatrix} = n,$$

meaning $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$. When vectors contained in a set are placed as columns into a matrix A , $\text{rank}(A)$ is the dimension of the vector space spanned by the vectors.

1.7 Determinants

1.7.1 Computing Determinants

- $n = 1$: $A = \begin{pmatrix} a \end{pmatrix}$, $\det A = a$
- $n = 2$: $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A = ad - bc$
- $n \geq 3$: The determinant can be computed by expanding the minor determinants (M_{ij}) of the elements (a_{ij}) of *any* row or column:

$$\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij})$$

1.7.2 Properties of Determinants

- $k \det A = \det B$ if a row or column of B is k times the corresponding row or column of A .
- $\det(kA) = k^n \det A$ if A is an $n \times n$ matrix.
- $\det A = 0$ if
 - A has a row or column of zeros,
 - A has a row or column that is a multiple of another row or column, or
 - A is not invertible.
- $\det B = (-1)^r \det A$ if B is obtained from A by swapping two rows or columns r times.
- $\det(A^T) = \det(A)$
- $\det A = \det U$ where U is the upper triangular matrix obtained from A by elementary row operations. $\det U$ is also the product of the diagonal entries of U .
- $\det(AB) = \det(A) \det(B)$

1.8 Linear Independence of Functions

A set of functions $\{f_0, f_1, \dots, f_n\}$ is linearly independent if the Wronstian determinant is non-zero:

$$W(f_0, f_1, \dots, f_n) = \begin{vmatrix} f_0 & f_1 & \dots & f_n \\ f'_0 & f'_1 & \dots & f'_n \\ \vdots & \vdots & \ddots & \vdots \\ f_0^{(n)} & f_1^{(n)} & \dots & f_n^{(n)} \end{vmatrix} \neq 0$$

1.9 Cramer's Rule

For a system of linear equations $A\mathbf{x} = \mathbf{b}$, the solution for x_i is given by

$$x_i = \frac{\det A_i}{\det A}$$

where A_i is the matrix obtained from A by replacing the i -th column with $\mathbf{b}$. Note that $\det A \neq 0$, which is necessary for a unique solution.

1.10 Matrix Inverse

The **matrix inverse** A^{-1} of A is such that $AA^{-1} = A^{-1}A = I$. It can be computed via

$$A^{-1} = \frac{1}{\det A} C^T,$$

where C is the matrix of cofactors of A . Remember that to find the cofactor C_{ij} , you need to find the determinant of the matrix A with the i -th row and j -th column removed, and then multiply it by $(-1)^{i+j}$.

The matrix inverse can also be computed by augmenting the matrix A with the identity matrix I and performing elementary row operations until the left side is the identity matrix. The right side will then be the inverse of A .

Not all matrices are invertible. A matrix is **invertible** iff its determinant is non-zero. In other words, A is invertible iff $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$.

1.11 Eigenvalues and Eigenvectors

For an $n \times n$ matrix A , an eigenvector $\mathbf{v}$ and its corresponding eigenvalue λ satisfy the equation

$$A\mathbf{v} = \lambda\mathbf{v},$$

where $\mathbf{v} \neq \mathbf{0}$. The eigenvalues are the roots of the characteristic equation

$$\det(A - \lambda I) = 0.$$

The eigenvectors are the non-zero solutions to

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

There may be up to n eigenvalues (including multiplicity) and up to n corresponding eigenvectors. Eigenvectors may be scaled by any non-zero scalar.

1.12 Similarity transformation

A square matrix A can be converted to a diagonal matrix D via a similarity transformation

$$D = C^{-1}AC,$$

where C is the matrix of eigenvectors and D is the diagonal matrix of eigenvalues. We say that C *diagonalizes* A or that A is **similar** to D . This is because D describes the same transformation as A but in a new coordinate system.

1.13 Orthogonal Matrices

An **orthogonal matrix** Q is a square matrix such that $Q^T Q = Q Q^T = I$. The columns (and rows) of an orthogonal matrix form an orthonormal basis, i.e., they are orthogonal to each other and have unit length. The action of Q on a vector preserves its length.

Note that $\det Q = \pm 1$. In particular:

- If $\det Q = 1$, then Q is a rotation matrix.
- If $\det Q = -1$, then Q is a reflection matrix.

1.14 Eigenvectors of Symmetric Matrices

If a square matrix A is symmetric, its eigenvectors are **orthogonal**. After normalizing them, the matrix C of orthonormal eigenvectors is an **orthogonal matrix**. The similarity transformation then becomes

$$D = C^{-1} A C = C^T A C,$$

which can be used to define a new *rotated orthogonal* coordinate system.

1.15 Rotation Matrices

A **rotation matrix** $R(\theta)$ performs a counter-clockwise rotation by angle θ and is given by

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Note that $R(\theta)$ is an **orthogonal matrix**. Also note that $R(\theta)^T$ represents a clockwise rotation by angle θ . Applying $R(\theta)$ to the standard basis vectors yields the rotated basis vectors.

An example of a rotation matrix in 3D is

$$R(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}.$$

Note that $R(\theta)$ is a passive transformation (corresponding to the $-\sin \theta$ in the bottom-left entry) about the y -axis (corresponding to the 2nd row matching I_3).

1.16 Active and Passive Transformations

A transformation is an **active transformation** if it transforms the *vector* $\mathbf{x}$ to a new vector $\mathbf{y}$ in the *same* coordinate system. For example, A below rotates the vector $\mathbf{x}$ counter-clockwise by angle θ to $\mathbf{y}$:

$$\mathbf{y} = A\mathbf{x}$$

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

A transformation is a **passive transformation** if it transforms the *coordinate system* (e.g., from unprimed to primed) *without* changing the vector $\mathbf{x}$ (i.e., $\mathbf{x} = \mathbf{x}'$). For example, A below rotates the coordinate system counter-clockwise by angle θ :

$$\mathbf{x}' = A\mathbf{x}$$

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

1.17 Application to Conics and Quadric Surfaces

1.17.1 Central Conics in Matrix Form

$$Ax^2 + 2Hxy + By^2 = K$$

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & H \\ H & B \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = K$$

$$\mathbf{x}^T M \mathbf{x} = K$$

1.17.2 Quadric Surfaces in Matrix Form

$$Ax^2 + By^2 + Cz^2 + 2Dxy + 2Exz + 2Fyz = K$$

$$\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} A & D & E \\ D & B & F \\ E & F & C \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = K$$

$$\mathbf{x}^T M \mathbf{x} = K$$

Instead of memorizing the formulas, remember that the matrix is symmetric and the diagonal entries are the coefficients of the squared terms, and the off-diagonal entries are half the coefficients of the cross terms. Also, the indices of the cross terms match the variables. For example, the term a_{12} corresponds to the cross term xy (x being 1 and y being 2).

1.17.3 Rotation via Principal Axes

Rotating a conic or quadric surface to its principal axes is useful for eliminating the cross terms (the non-diagonal elements in matrix form), thereby making it easier to analyze.

$$\begin{pmatrix} x' & y' \end{pmatrix} C^{-1} M C \begin{pmatrix} x' \\ y' \end{pmatrix} = K$$

$$\begin{pmatrix} x' & y' \end{pmatrix} D \begin{pmatrix} x' \\ y' \end{pmatrix} = K$$

where

- M is the matrix of the coefficients of the conic or quadratic equation,
- D is the diagonal matrix of the eigenvalues of M ,
- C is the matrix of the eigenvectors of M .

Note that if M is symmetric, then C is orthogonal, i.e., $C^{-1} = C^T$.

1.18 Multi-Mass-Spring Systems

Consider a system of 2 masses m_1 and m_2 connected by 3 springs of spring constants k_1 , k_2 , and k_3 . The system is one-dimensional and is in the form spring-mass-spring-mass-spring, in which the first and last springs are connected to fixed walls on the left and right, respectively. The objective is to find the equations of motion. The general steps are:

1. Recall that a Hookian spring force is given by

$$\mathbf{F} = -k\mathbf{x}$$

where k is the spring constant and $\mathbf{x}$ is the displacement of the spring from its equilibrium position.

2. Recall that

$$\mathbf{F} = -\nabla U$$

where U is the potential energy of the system. In one dimension:

$$F = -\frac{\partial U}{\partial x}.$$

3. If x_1 and x_2 are the displacements of the first and second masses from their equilibrium positions, respectively, then the middle spring's compression/extension is given by $x_2 - x_1$.
4. Solve for U in terms of k_1, k_2, k_3, x_1 , and x_2 .
5. Recall that

$$-\frac{\partial U}{\partial x} = F = ma = m\ddot{x}.$$

6. Since this involves harmonic motion, assume the form

$$x_n(t) = A_n e^{i\omega t},$$

so

$$\ddot{x}_n = -\omega^2 x_n.$$

7. Designate

$$\lambda_n = \frac{m\omega_n^2}{k},$$

where, in fact, λ is the eigenvalue of the linear system.

8. Solve for the eigenvalues and eigenvectors of the system. The eigenvalues give the **characteristic frequencies** (when solving for ω_n) of the system, and the eigenvectors give the **characteristic modes of vibration** of the system. For this particular example, if both elements in an eigenvector are of the same sign, then the mode of vibration would be $\rightarrow\rightarrow$ and $\leftarrow\leftarrow$. If the elements are of opposite signs, then the mode of vibration at the corresponding frequency would be $\rightarrow\leftarrow$ and $\leftarrow\rightarrow$.

2 Vector Analysis

2.1 Scalar Fields

A scalar field $f(\mathbf{x})$ is a function that maps each point $\mathbf{x} = x_1\hat{x}_1 + x_2\hat{x}_2 + \dots + x_n\hat{x}_n$ to a scalar: $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Some examples include temperature and pressure as a function of position.

2.2 Vector Fields

A vector field $\mathbf{F}(\mathbf{x}) = F_1(\mathbf{x})\hat{x}_1 + F_2(\mathbf{x})\hat{x}_2 + \cdots + F_m(\mathbf{x})\hat{x}_m$ is a function that maps each point $\mathbf{x} = x_1\hat{x}_1 + x_2\hat{x}_2 + \cdots + x_n\hat{x}_n$ to a vector: $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Note that each $F_i(\mathbf{x})$ is a scalar field. Some examples include velocity and acceleration as a function of position.

2.3 Dot Product

Also known as the scalar product, the dot product takes two vectors and returns a scalar.

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= |\mathbf{A}||\mathbf{B}| \cos \theta \\ &= A_x B_x + A_y B_y + A_z B_z\end{aligned}$$

Properties:

- Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- $\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2$
- $\mathbf{A} \cdot \mathbf{B} = 0 \iff \mathbf{A} \perp \mathbf{B}$
- *Not* associative

A useful identity involving the dot product is

$$\mathbf{A} \cdot d\mathbf{A} = |\mathbf{A}|d|\mathbf{A}|$$

2.4 Projections

The scalar projection of $\mathbf{A}$ onto $\mathbf{B}$ is given by

$$\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|} = |\mathbf{A}| \cos \theta$$

The vector projection of $\mathbf{A}$ onto $\mathbf{B}$ is given by

$$\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|^2} \mathbf{B} = (\mathbf{A} \cdot \mathbf{u}) \mathbf{u}$$

2.5 Gram-Schmidt Orthonormalization

Given any two vectors $\mathbf{A}$ and $\mathbf{B}$, we can find an orthonormal basis $\mathbf{v}_1$ and $\mathbf{v}_2$ for the plane spanned by $\mathbf{A}$ and $\mathbf{B}$ as

$$\begin{aligned}\mathbf{v}_1 &= \frac{\mathbf{A}}{|\mathbf{A}|} = \mathbf{u} \\ \mathbf{v}_2 &= \frac{\mathbf{B} - (\mathbf{B} \cdot \mathbf{u})\mathbf{u}}{|\mathbf{B} - (\mathbf{B} \cdot \mathbf{u})\mathbf{u}|}\end{aligned}$$

2.6 Equation of a Plane

$$\begin{aligned}\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) &= 0 \\ n_x(x - x_0) + n_y(y - y_0) + n_z(z - z_0) &= 0\end{aligned}$$

where

- $\mathbf{r} = (x, y, z)$ is a vector to any point on the plane,
- $\mathbf{n}$ is the normal vector to the plane, and
- $\mathbf{r}_0 = (x_0, y_0, z_0)$ is a known point on the plane.

2.7 Cross Product

Also known as the vector product, the cross product takes two vectors and returns a vector.

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= (|\mathbf{A}||\mathbf{B}|\sin\theta)\mathbf{u} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}\end{aligned}$$

where $\mathbf{u}$ is the unit vector perpendicular to both $\mathbf{A}$ and $\mathbf{B}$. Use the **right-hand rule** to determine the correct direction of $\mathbf{u}$. Geometrically, the magnitude of the cross product represents the area of the parallelogram created by $\mathbf{A}$ and $\mathbf{B}$.

Properties:

- Anti-commutative: $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$
- Distributive: $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$
- $\mathbf{A} \times \mathbf{B} = \mathbf{0} \iff \mathbf{A} \parallel \mathbf{B}$
 - Thus, $\mathbf{A} \times \mathbf{A} = \mathbf{0}$.

2.8 Equation of a 3D Line

$$\begin{aligned}\mathbf{r} &= \mathbf{r}_0 + t\mathbf{u} \\ \mathbf{r} \times \mathbf{u} &= \mathbf{r}_0 \times \mathbf{u}\end{aligned}$$

where

- $\mathbf{r} = (x, y, z)$ is a vector to any point on the line,
- $\mathbf{r}_0 = (x_0, y_0, z_0)$ is a known point on the line,
- $\mathbf{u}$ is a unit vector in the direction of the line, and
- t is a scalar parameter.

2.9 Triple Scalar Product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

Performing an even number of swaps on the determinant results in the same answer (which is equivalent to performing **cyclic permutations** such as $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$), whereas performing an odd number of swaps simply results in an answer of the opposite sign.

Geometrically, the triple scalar product represents the volume of a parallelepiped formed by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. More specifically, any two vectors can be used to form the base of the parallelepiped (e.g., $\mathbf{A}$ and $\mathbf{B}$), and the height is the third vector's ($\mathbf{C}$) projection onto the normal vector of the base ($\mathbf{A} \times \mathbf{B}$). Furthermore, $\frac{1}{6}$ of the volume of the parallelepiped is the volume of the tetrahedron formed by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$.

2.10 Triple Vector Product

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

To remember this formula (beyond the mnemonic BAC-CAB), the right hand side is a linear combination of the vectors in the parentheses on the left hand side. The coefficients are the outside vector dotted with each of the inside vectors. Both terms have $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ exactly once. Finally, the positive term on the right hand side belongs to the vector in the middle on the left hand side ($\mathbf{B}$ in this case).

$\mathbf{B} \times \mathbf{C}$ is perpendicular to the plane spanned by $\mathbf{B}$ and $\mathbf{C}$, but $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is back on that plane.

2.11 Polar Coordinates

In rectangular coordinates, the basis vectors $\mathbf{i}$ and $\mathbf{j}$ are constant for all points, but in polar coordinates, the basis vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ are *not* constant.

$$\begin{aligned}\mathbf{e}_r &= \cos \theta \mathbf{i} + \sin \theta \mathbf{j} \\ \mathbf{e}_\theta &= -\sin \theta \mathbf{i} + \cos \theta \mathbf{j} \\ \frac{d\mathbf{e}_r}{dt} &= \mathbf{e}_\theta \frac{d\theta}{dt} \\ \frac{d\mathbf{e}_\theta}{dt} &= -\mathbf{e}_r \frac{d\theta}{dt}\end{aligned}$$

To find the derivative of a vector in polar coordinates, the chain rule must be remembered. In rectangular coordinates, the derivative of a basis vector is always zero, leading to the result being the simple derivative of the components, but this is *not* the case in polar coordinates.

2.12 Spherical Coordinates

$$\begin{aligned}\mathbf{e}_\rho &= \cos \theta \sin \phi \mathbf{i} + \sin \theta \sin \phi \mathbf{j} + \cos \phi \mathbf{k} \\ \mathbf{e}_\theta &= -\sin \theta \mathbf{i} + \cos \theta \mathbf{j} \\ \mathbf{e}_\phi &= \cos \theta \cos \phi \mathbf{i} + \sin \theta \cos \phi \mathbf{j} - \sin \phi \mathbf{k}\end{aligned}$$

Note that θ and ϕ can be swapped, but the convention is the class is to assign θ to the xy -plane and ϕ to the z -axis.

2.13 Gradient

The gradient represents the direction and magnitude of the greatest rate of increase of a scalar field. The vector opposite of the gradient is the direction of the greatest rate of decrease. The vectors perpendicular to the gradient represent no change in the scalar field.

In rectangular coordinates:

$$\nabla f = \text{grad } f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

where f is a scalar field. Note that ∇f

- is a vector field.
- points in the direction of the greatest rate of increase of f .

- is perpendicular to the surface $f = c$ for any constant c .

In cylindrical coordinates:

$$\nabla f = \text{grad } f = \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{\partial f}{\partial z} \mathbf{e}_z$$

In spherical coordinates:

$$\nabla f = \text{grad } f = \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho \sin \phi} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi$$

2.14 Directional Derivative

$$\frac{df}{ds} = \nabla f \cdot \mathbf{u}$$

where $\mathbf{u}$ is the unit vector in the direction of differentiation.

Special cases in rectangular coordinates:

- $\mathbf{u} = \mathbf{i} \implies \frac{df}{ds} = \frac{\partial f}{\partial x}$
- $\mathbf{u} = \mathbf{j} \implies \frac{df}{ds} = \frac{\partial f}{\partial y}$
- $\mathbf{u} = \mathbf{k} \implies \frac{df}{ds} = \frac{\partial f}{\partial z}$

These are simply the coefficients in front of the basis vectors from the definition of the gradient. The special cases for cylindrical and spherical coordinates can be found similarly.

2.15 Divergence

In rectangular coordinates:

$$\nabla \cdot \mathbf{V} = \text{div } \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

where $\mathbf{V}$ is a vector field. Note that $\nabla \cdot \mathbf{V}$ is a scalar field.

In cylindrical coordinates:

$$\nabla \cdot \mathbf{V} = \frac{1}{r} \frac{\partial}{\partial r}(r V_r) + \frac{1}{r} \frac{\partial}{\partial \theta} V_\theta + \frac{\partial}{\partial z} V_z$$

In spherical coordinates:

$$\nabla \cdot \mathbf{V} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 V_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(V_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial V_\phi}{\partial \phi}$$

2.16 Curl

In rectangular coordinates:

$$\begin{aligned}\nabla \times \mathbf{V} &= \text{curl } \mathbf{V} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} \\ &= \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \mathbf{i} + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) \mathbf{j} + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) \mathbf{k}\end{aligned}$$

where $\mathbf{V}$ is a vector field. Note that $\nabla \times \mathbf{V}$ is a vector field.

2.17 Laplacian

In rectangular coordinates:

$$\begin{aligned}\nabla^2 f &= \nabla \cdot (\nabla f) = \text{div grad } f \\ &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}\end{aligned}$$

In cylindrical coordinates:

$$\nabla^2 f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}$$

In spherical coordinates:

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$$

If f is only a function of r , then the spherical Laplacian simplifies to

$$\nabla^2 f = \frac{d^2 f}{dr^2} + \frac{2}{r} \frac{df}{dr} = \frac{1}{r} \frac{d^2}{dr^2} (rf).$$

2.18 Vector Operator Properties

- A field is **irrotational** iff $\nabla \times \mathbf{F} = \mathbf{0}$
 - Gradient fields are irrotational: $\nabla \times (\nabla f) = \mathbf{0}$
- A field is **incompressible** (or **solenoidal**) iff $\nabla \cdot \mathbf{F} = 0$
 - $\nabla \cdot (\nabla \times \mathbf{F}) = 0$
- Linearity:

- $\nabla(f + g) = \nabla f + \nabla g$ and $\nabla(cf) = c\nabla f$
- $\nabla \cdot (\mathbf{F} + \mathbf{G}) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}$ and $\nabla \cdot (c\mathbf{F}) = c(\nabla \cdot \mathbf{F})$
- $\nabla \times (\mathbf{F} + \mathbf{G}) = \nabla \times \mathbf{F} + \nabla \times \mathbf{G}$ and $\nabla \times (c\mathbf{F}) = c(\nabla \times \mathbf{F})$

- Product rule:

- $\nabla(fg) = f\nabla g + g\nabla f$
- $\nabla \cdot (f\mathbf{F}) = f(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot (\nabla f)$
- $\nabla \times (f\mathbf{F}) = f(\nabla \times \mathbf{F}) + (\nabla f) \times \mathbf{F}$

- Gradient quotient rule:

$$\nabla \left(\frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}$$

- $\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$
- $\nabla^2(fg) = f \nabla^2 g + g \nabla^2 f + 2(\nabla f \cdot \nabla g)$
- Using properties above: $\nabla \cdot (\nabla f \times \nabla g) = 0$
- $\nabla \cdot (f\nabla g - g\nabla f) = f \nabla^2 g - g \nabla^2 f$
- Using the triple vector product: $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$
 - Note that this property also defines $\nabla^2 \mathbf{F}$.
- $\nabla \times (\mathbf{F} \times \mathbf{G}) = \mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}$

2.19 Unit Tangent and Normal Vectors

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \mathbf{r}'(s) = \frac{x'(t)\mathbf{i} + y'(t)\mathbf{j}}{\sqrt{x'(t)^2 + y'(t)^2}}$$

$$\mathbf{N} = \frac{y'(t)\mathbf{i} - x'(t)\mathbf{j}}{\sqrt{x'(t)^2 + y'(t)^2}}$$

where

- $\mathbf{r}(t) : \mathbb{R}^1 \rightarrow \mathbb{R}^n$ is a parameterization of a curve C ,
- $x'(t)dt = dx$ and $y'(t)dt = dy$, and
- $\mathbf{N}$ points outward from the region enclosed by C .

2.20 Line Integrals and Circulation

For a scalar field f in 1D:

$$\int_C f \, ds = \int_a^b f \, dt$$

where C is the set of all points $t \in [a, b]$, and ds is a differential of arc length.

For a scalar field f in 2D and beyond:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt = \int_{-C} f \, ds$$

where $ds = |\mathbf{r}'(t)| \, dt$ and the direction of integration does not matter.

For a vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ in 2D:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt = \int_C \mathbf{F} \cdot \mathbf{T} \, ds = \int_C P \, dx + Q \, dy = - \int_{-C} \mathbf{F} \cdot d\mathbf{r}$$

Note that the direction of integration matters for vector fields, unlike scalar fields. Also, $\mathbf{F} \cdot \mathbf{T} = |\mathbf{F}| \cos \theta$ represents the amount of $\mathbf{F}$ in the direction of C . Thus, the above integral represents the **circulation** of $\mathbf{F}$ along C , and $\mathbf{F} \cdot \mathbf{T}$ is the circulation or flow per unit length along C .

2.21 Flux

$$\oint_C \mathbf{F} \cdot d\mathbf{n} = \oint_C \mathbf{F} \cdot \mathbf{N} \, ds = \oint_C \mathbf{F}(\mathbf{r}(t)) \cdot (dy\mathbf{i} - dx\mathbf{j}) = \oint_C P \, dy - Q \, dx$$

where $d\mathbf{n} = \mathbf{N} \, ds$ pertains to a region enclosed by C .

2.22 Conservative Vector Fields, Scalar Potentials, and Exact Differentials

A field $\mathbf{V}$ is **conservative** iff $\nabla \times \mathbf{V} = \mathbf{0}$. Such a field is also called a **gradient field**. If $\mathbf{V}$ is a gradient field, then there exists a **scalar potential** ϕ such that $\mathbf{V} = \nabla \phi$. (In physics, $\mathbf{V} = -\nabla \phi$ is used instead.)

$$\begin{aligned} \oint_C \nabla \phi \cdot d\mathbf{r} &= 0 \\ \int_C \nabla \phi \cdot d\mathbf{r} &= \phi(\mathbf{r}(b)) - \phi(\mathbf{r}(a)) \end{aligned}$$

Note the difference between the two integrals. The first is a closed curve, while the second is an open curve. A physical example is potential difference: it is zero around a closed curve, and it is simply the difference between the values at the two endpoints for an open curve.

If $\mathbf{F} = \nabla W$ (i.e., $\mathbf{F}$ is conservative), then $dW = dW_x dx + dW_y dy + dW_z dz$ is an exact differential, which means that

$$\frac{\partial dW_x}{\partial y} = \frac{\partial dW_y}{\partial x}$$

for all pairs of variables. The idea of conservative fields and exact differentials are equivalent to one another.

2.23 Green's Theorem

$$\iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_{\partial A} P dx + Q dy$$

where P and Q are continuous and have continuous first partial derivatives on area A . The line integral is counterclockwise around ∂A , the boundary of A . Going in the clockwise direction would introduce a negative sign. Green's theorem is convenient because it allows one to choose the easier of the two integrals to evaluate.

2.24 Tangent and Outward Normal Vectors

$$\begin{aligned} d\mathbf{r} &= dx \mathbf{i} + dy \mathbf{j} && \text{(tangent)} \\ \mathbf{n} ds &= dy \mathbf{i} - dx \mathbf{j} && \text{(outward normal)} \end{aligned}$$

where $\mathbf{n}$ is a unit vector and $ds = \sqrt{dx^2 + dy^2}$.

2.25 Divergence Theorem

$$\begin{aligned} \iint_A \nabla \cdot \mathbf{V} dx dy &= \int_{\partial A} \mathbf{V} \cdot \mathbf{n} ds && \text{(2-dimensional)} \\ \iiint_{\tau} \nabla \cdot \mathbf{V} d\tau &= \iint_{\partial \tau} \mathbf{V} \cdot \mathbf{n} d\sigma && \text{(3-dimensional)} \end{aligned}$$

where τ represents volume and $d\sigma$ represents an element of surface area.

2.26 Stokes' Theorem

$$\iint_{\sigma} (\nabla \times \mathbf{V}) \cdot \mathbf{n} \, d\sigma = \oint_{\partial\sigma} \mathbf{V} \cdot d\mathbf{r}$$

where σ is an open surface and $\partial\sigma$ is its boundary curve. Stokes' theorem is the 3D generalization of Green's theorem.

To calculate $\mathbf{n} \, d\sigma$ for a surface, it is useful to parameterize the flattened surface defined by $\mathbf{r}(u, v)$, resulting in

$$\mathbf{n} \, d\sigma = (\mathbf{r}_u \times \mathbf{r}_v) \, du \, dv.$$

3 Partial Differential Equations

3.1 Introduction

Partial differential equations (PDEs) involve partial derivatives of a function with respect to multiple independent variables. They are widely used in physics and engineering to model various phenomena. The most common PDEs are:

$$\nabla^2 u = 0 \quad \text{(Laplace's equation)}$$

$$\nabla^2 u = f(x, y, z) \quad \text{(Poisson's equation)}$$

$$\nabla^2 u = \frac{1}{\alpha^2} \frac{\partial u}{\partial t} \quad \text{(Heat equation)}$$

$$\nabla^2 u = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} \quad \text{(Wave equation)}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi = i\hbar \frac{\partial}{\partial t} \Psi \quad \text{(Schrödinger's equation)}$$

These often form a class of problems called **eigenvalue boundary value problems**. Note that all of these have a second derivative with respect to space, leading to the solution often being a sum of sines and cosines.

3.2 Common Problems

3.2.1 Heat Flow in a Semi-Infinite Channel

Given a channel of width L and infinite height, with the bottom held at temperature T_0 , the steady-state temperature distribution is given by

$$\nabla^2 T = 0, \text{ for } 0 < x < L \text{ and } y > 0$$

Possible boundary conditions:

$$T(x, 0) = T_0 \quad (1)$$

$$T(0, y) = 0 \text{ or } T_x(0, y) = 0 \quad (2)$$

$$T(L, y) = 0 \text{ or } T_x(L, y) = 0 \quad (3)$$

$$T(x, y) \rightarrow 0 \text{ as } y \rightarrow \infty$$

The T_x and T_y boundary conditions imply insulation or zero flux at the boundaries. Usually, boundary conditions (2) and (3) are applied to the individual solutions to find the eigenvalues and eigenfunctions, and boundary condition (1) is applied to the general solution to find the series coefficients.

The y direction often has an exponential decay (e.g., $e^{-\lambda_n y}$) to fit the boundary condition $T(x, y) \rightarrow 0$ as $y \rightarrow \infty$.

3.2.2 Heat Flow Infinitely Thin Rod

Given an infinitely thin rod from $x = 0$ to $x = L$ that is insulated above and below (i.e., heat can only travel in the x direction), the heat flow is given by

$$u_t = \alpha^2 u_{xx}, \text{ for } 0 < x < L \text{ and } t > 0$$

Possible boundary conditions:

$$u(0, t) = 0 \text{ or } u_x(0, t) = 0$$

$$u(L, t) = 0 \text{ or } u_x(L, t) = 0$$

Initial condition:

$$u(x, 0) = f(x)$$

If the rod were instead a ring from $-L$ to L , there would be periodic boundary conditions:

$$u(-L, t) = u(L, t)$$

$$u_x(-L, t) = u_x(L, t)$$

The solution for the function of t often involves exponential decay (e.g., $e^{-\alpha^2 \lambda_n t}$), which makes sense since heat dissipates over time.

3.2.3 Wave on an Infinitely Thin String

Common wave notation:

$$\begin{aligned}
\nu &= \text{frequency (sec}^{-1}\text{)} \\
\lambda &= \text{wavelength (m)} \\
c &= \nu\lambda = \text{wave speed (m/sec)} \\
\omega &= 2\pi\nu = \text{angular frequency (rad/sec)} \\
k &= \frac{2\pi}{\lambda} = \frac{2\pi\nu}{c} = \frac{\omega}{c} = \text{wave number (rad/m)}
\end{aligned}$$

Given an infinitely thin string of length L , the wave equation is given by

$$u_{tt} = c^2 u_{xx}, \text{ for } 0 < x < L \text{ and } t > 0,$$

where $u(x, t)$ is the vertical displacement of the string at position x and time t , and c is the wave speed.

Possible boundary conditions:

$$\begin{aligned}
u(0, t) = 0 = u(L, t) & \quad \text{(fixed ends)} \\
u_x(0, t) = 0 = u_x(L, t) & \quad \text{(free ends)}
\end{aligned}$$

These may be mixed, e.g. one end fixed and the other free.

Initial conditions:

$$\begin{aligned}
u(x, 0) &= f(x) \\
u_t(x, 0) &= g(x)
\end{aligned}$$

Because the PDE contains a second partial derivative with respect to time, there needs to be two initial conditions (namely one for displacement, and one for velocity) to find the solutions.

The solution to a generic wave equation is

$$u(x, t) = f(x + ct) + g(x - ct)$$

where $f(x + ct)$ is a leftward moving wave and $g(x - ct)$ is a rightward moving wave. This is called **d'Alembert's solution**.

A common initial condition is a plucked string, where the string is pulled up at a certain point and released from rest. This corresponds to the initial conditions

$$\begin{aligned}
u(x, 0) &= f(x) \\
u_t(x, 0) &= 0
\end{aligned}$$

where $f(x)$ is a triangular function.

For a wave fixed at both ends, the lowest frequency at which a standing wave can occur is called the **fundamental frequency** (or **first harmonic**), given by

$$\nu_1 = \frac{c}{2L},$$

and each subsequent frequency is an integer multiple of the fundamental frequency:

$$\nu_n = n\nu_1 = \frac{nc}{2L}, \text{ for } n = 1, 2, 3, \dots$$

The second harmonic (or **first overtone**) is ν_2 ; the third harmonic (or second overtone) is ν_3 , and so on.

3.2.4 Schrödinger's Equation

This class does not focus on Schrödinger's equation, but it is worth noting that there is often an oscillatory solution (unlike in the heat equation, which has a decaying solution) for the time-dependent part of Schrödinger's equation as well as the space-dependent part, owing to the existence of i in the PDE:

$$i\hbar \frac{\partial}{\partial t} \Psi = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi$$

3.2.5 Steady-State in an Annulus

Given an annulus with inner radius a and outer radius b , the steady-state temperature distribution is given by

$$\nabla^2 u = 0, \text{ for } a < r < b.$$

Separation of variables in polar coordinates:

$$u(r, \theta) = R(r)\Theta(\theta)$$

Implied periodic angular boundary conditions:

$$\begin{aligned} u(r, \theta) &= u(r, \theta + 2\pi) \\ u_\theta(r, \theta) &= u_\theta(r, \theta + 2\pi) \end{aligned}$$

Possible radial boundary conditions:

$$\begin{aligned} u(a, \theta) &= f(\theta) \\ u(b, \theta) &= g(\theta) \end{aligned}$$

ODEs:

$$\begin{aligned}\Theta'' + \lambda\Theta &= 0 \\ r^2 R'' + rR' - \lambda R &= 0 \Leftrightarrow r(rR')' = \lambda R\end{aligned}$$

It is easier to apply the periodic boundary conditions first to solve for $\Theta(\theta)$, yielding sines and cosines. Then, apply the radial boundary conditions to solve for $R(r)$, making sure to check $\lambda = 0$ as well. These yield polynomials in r . After using the principle of superposition, the general solution mimics the Fourier series with the coefficients being slightly different as determined by the boundary condition.

3.3 Eigenvalues and Eigenfunctions

In many PDEs, the solution is of the form

$$u(x, y) = \sum_{n=1}^{\infty} c_n X_n(x) Y_n(y)$$

where $X_n(x)$ and $Y_n(y)$ are the **eigenfunctions** and the λ_n contained within them are the **eigenvalues**. For example, if $X_n(x) = \sin(\frac{n\pi x}{L})$, then the eigenvalues are $\lambda_n = (\frac{n\pi}{L})^2$, which are paired one-to-one for each n .

The eigenfunctions form an orthogonal basis, e.g.

$$\{1, X_1, X_2, \dots, X_n, \dots\}$$

analogous to standard basis vectors. Note: the 1 is included for the case where $\lambda = 0$.

3.4 General Strategy

1. Assume a separable form, e.g. $u = X(x)Y(y)$.
2. Differentiate u as needed and substitute into the original PDE.
3. Divide by XY and set equal to a constant, e.g. $-\lambda$.
4. Solve the resulting ODEs.
5. Apply the boundary conditions to find the **eigenvalues** and their corresponding **eigenfunctions** $X_n(x)$ or $Y_n(y)$.
 - (a) Enforce a non-trivial solution to get discrete eigenvalues, e.g.

$$\sin(\sqrt{\lambda}L) = 0 \implies \sqrt{\lambda}L = n\pi \implies \lambda = \left(\frac{n\pi}{L}\right)^2$$

- (b) Make sure to test if $\lambda = 0$ is an eigenvalue.

- Construct the general solution as a linear combination of the product of the individual solutions, forming an infinite series:

$$u(x, y) = \sum_{n=1}^{\infty} c_n X_n(x) Y_n(y)$$

- Apply the initial conditions or the rest of the boundary conditions (which cannot satisfy the individual solutions) to the general solution to find the series coefficients c_n , e.g.

$$c_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

if the eigenfunction is sine. In general, the equation for c_n is the inner product of the eigenfunction and the initial condition, normalized by the norm of the eigenfunction (e.g. the $2/L$ in the example above).

- For non-homogeneous problems, first solve the homogeneous problem to find the general solution. For example, if the PDE is

$$u(x, t) = v(x, t) + u_f(x),$$

where $u_f(x)$ is the **steady-state solution** and $v(x, t)$ is the **transient solution**, solve for $v(x, t)$ with homogenous boundary conditions and initial condition $v(x, 0) = u(x, 0) - u_f(x)$.

- Apply common simplifications, such as $\cos(n\pi) = (-1)^n$, $\sin(n\pi) = 0$, and considering odd and even cases (one of which is sometimes 0). If only odd terms remain, use $2n - 1$ instead of n (or $2n + 1$ with starting index $n = 0$) in the general solution. If only even terms remain, use $2n$ instead of n in the general solution.

3.5 Separation of Variables

Take example of $\nabla^2 u = 0$. The first step of solving a partial differential equation (PDE) is to assume that it is **separable**:

$$u = X(x)Y(y)$$

Differentiate u as needed and substitute into the original PDE:

$$X''Y + XY'' = 0$$

If the PDE is separable, dividing by XY will separate the variables:

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

The $-\lambda$ is called the **separation constant**, and it is usually chosen to be negative to fit the boundary conditions.

3.6 Solutions to Common ODEs

$$\begin{array}{l|l} x' + \lambda x = 0 & x = c_1 e^{-\lambda x} \\ x'' = 0 & x = c_1 + c_2 x \\ x'' + \lambda x = 0 & x = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x) \\ x'' - \lambda x = 0 & x = c_1 e^{\sqrt{\lambda}x} + c_2 e^{-\sqrt{\lambda}x} \\ r^2 x'' + r x' - \lambda x = 0 & x = c_1 r^{\sqrt{\lambda}} + c_2 r^{-\sqrt{\lambda}} \end{array}$$

4 Fourier Series

4.1 Introduction

Given a $2L$ -periodic function $f(x)$, its Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

where

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \end{aligned}$$

Note that the a_n formula applies for a_0 too with $\cos(0) = 1$. The constant $a_0/2$ is also the average value of $f(x)$.

The Fourier series shows that a $f(x)$ can be represented as a sum of even (cosine) and odd (sine) functions. Thus, the following is true:

- $f(x)$ is even $\implies b_n = 0$
- $f(x)$ is odd $\implies a_n = 0$

The constants $a_0/2$, a_n , and b_n are the components of $f(x)$ in the directions of the basis functions

$$\left\{ 1, \cos\left(\frac{\pi x}{L}\right), \sin\left(\frac{\pi x}{L}\right), \cos\left(\frac{2\pi x}{L}\right), \sin\left(\frac{2\pi x}{L}\right), \dots \right\}.$$

The convergence of the Fourier series relies heavily on the coefficients a_n and b_n since the trigonometric functions simply oscillate between -1 and 1 .

4.2 Complex Form

The Fourier series can be equivalently written in complex form as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{n\pi x}{L}}$$

where

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i \frac{n\pi x}{L}} dx.$$

The i and $-i$ in $f(x)$ and c_n can be swapped without changing the result.

Note that $c_n = c_{-n}^*$ iff $f(x)$ is real-valued. Similar to the non-complex form, each c_n is the component of $f(x)$ in the direction of the basis function $e^{i \frac{n\pi x}{L}}$.

4.3 Parseval's Relation

$$\frac{1}{2L} \int_{-L}^L |f(x)|^2 dx = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \sum_{n=-\infty}^{\infty} |c_n|^2$$

where $|z|^2$ for a complex value or function z is zz^* , so

$$|c_n|^2 = c_n c_n^* = \frac{a_n - ib_n}{2} \frac{a_n + ib_n}{2} = \frac{a_n^2 + b_n^2}{4}. \quad (1)$$

4.4 Theorem of Dirichlet

If

- $f(x)$ is periodic with period $2L$,
- $f(x)$ is single-valued on $[-L, L]$,
- $f(x)$ has a finite number of maximum and minimum values on $[-L, L]$,
- $f(x)$ has a finite number of discontinuities on $[-L, L]$, and
- $\int_{-L}^L |f(x)| dx < \infty$,

then the Fourier series of $f(x)$ converges to $\frac{f(x^+) + f(x^-)}{2}$ for all x . That is, at points where $f(x)$ is continuous, the Fourier series simply converges to that value, and at jumps, the series converges to the midpoint of the jump (including at $x = \pm L$).

Some functions that do not satisfy the Dirichlet conditions include:

- Not single-valued: $x^2 + y^2 = L^2$
- Infinite maxima and minima: $f(x) = \sin(1/x)$
- Infinite discontinuities:

$$f(x) = \begin{cases} 1 & \text{for } x > 0 \text{ in } \sin(1/x) \\ -1 & \text{for } x < 0 \text{ in } \sin(1/x) \end{cases}$$

- Not absolutely integrable: $f(x) = 1/x$

Note that the converse of the theorem of Dirichlet is *not* true. That is, a function that does not satisfy the Dirichlet conditions may still have a convergent Fourier series. In fact, $f(x) = \sin(1/x)$ is such a function.

5 Fourier Transform

5.1 Definition

$$\mathcal{F}\{f(x)\} = F(\alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x)e^{-i\alpha x} dx$$

$$\mathcal{F}^{-1}\{F(\alpha)\} = f(x) = \int_{-\infty}^{\infty} F(\alpha)e^{i\alpha x} d\alpha$$

where $f(x) \rightarrow 0$ as $x \rightarrow \pm\infty$.

Other definitions exist, such as that obtained by swapping the $1/(2\pi)$ factor between the two equations or using $1/\sqrt{2\pi}$ for both.

The Fourier transform essentially takes the limit $L \rightarrow \infty$ on the Fourier series, lifting the constraint of a $2L$ -periodic function. That is, the Fourier transform is a generalization of the Fourier series and can be used on non-periodic functions as well. This is useful for problems such as the wave equation with an infinitely long string. Furthermore, the Fourier transform is inherently continuous due to α whereas the Fourier series is discrete due to n .

5.2 Properties

5.2.1 Differentiation

The Fourier transform turns differentiation into multiplication:

$$\mathcal{F}\{f'(x)\} = i\alpha F(\alpha) \quad (1\text{st derivative})$$

$$\mathcal{F}\{f^{(n)}(x)\} = (i\alpha)^n F(\alpha) \quad (n\text{th derivative})$$

5.2.2 Integration

The Fourier transform turns integration into division:

$$\mathcal{F} \left\{ \int_{-\infty}^x f(s) ds \right\} = \frac{F(\alpha)}{i\alpha} + \pi F(0)\delta(\alpha)$$

The $\pi F(0)\delta(\alpha)$ term represents a constant spike at $\alpha = 0$.

5.3 Solving the Wave Equation for an Infinite String

The Fourier transform can be used to solve the wave equation for an infinite string, which is given by

$$u_{tt} - c^2 u_{xx} = 0, \text{ for } -\infty < x < \infty, t > 0$$

with initial conditions

$$u(x, 0) = f(x), u_t(x, 0) = g(x)$$

where $u(x, t)$ is the displacement of the string at position x and time t , and c is the wave speed.

Represent the solution $u(x, t)$ as an inverse Fourier transform:

$$u(x, t) = \int_{-\infty}^{\infty} U(\alpha, t) e^{i\alpha x} d\alpha$$

Differentiate with respect to t and x :

$$\begin{aligned} u_{tt} &= \int_{-\infty}^{\infty} U''(\alpha, t) e^{i\alpha x} d\alpha \\ u_{xx} &= - \int_{-\infty}^{\infty} \alpha^2 U(\alpha, t) e^{i\alpha x} d\alpha \end{aligned}$$

Note that $U(\alpha, t)$ is strictly in terms of a single independent variable t because α is just a dummy variable.

Substitute these into the wave equation to yield a second-order ODE in terms of t :

$$\int_{-\infty}^{\infty} \underbrace{[U''(\alpha, t) + \alpha^2 c^2 U(\alpha, t)]}_{=0} e^{i\alpha x} d\alpha = 0$$

$$U''(\alpha, t) + \alpha^2 c^2 U(\alpha, t) = 0$$

The initial conditions are equivalently

$$\begin{aligned}U(\alpha, 0) &= F(\alpha) \\U'(\alpha, 0) &= G(\alpha)\end{aligned}$$

Thus, the general solution for $U(\alpha, t)$ is

$$\begin{aligned}U(\alpha, t) &= c_1(\alpha) \cos(c\alpha t) + c_2(\alpha) \sin(c\alpha t) \\U(\alpha, t) &= A(\alpha) \cos(c\alpha t) + B(\alpha) \frac{\sin(c\alpha t)}{c\alpha}\end{aligned}$$

where $A(\alpha) = c_1(\alpha)$ and $B(\alpha) = c\alpha c_2(\alpha)$ are arbitrary functions of α that are determined by the initial conditions.

Applying the initial conditions:

$$\begin{aligned}U(\alpha, 0) &= F(\alpha) = A(\alpha) \\U'(\alpha, 0) &= G(\alpha) = B(\alpha)\end{aligned}$$

Thus, the solution for $U(\alpha, t)$ is

$$U(\alpha, t) = F(\alpha) \cos(c\alpha t) + G(\alpha) \frac{\sin(c\alpha t)}{c\alpha}$$

Set $g(x) = 0$ (thus $G(\alpha) = 0$) to find the solution for an initial displacement $f(x)$ with zero initial velocity:

$$\begin{aligned}u(x, t) &= \frac{1}{2} \int_{-\infty}^{\infty} F(\alpha) e^{i\alpha(x+ct)} + F(\alpha) e^{i\alpha(x-ct)} d\alpha \\&= \frac{1}{2} f(x+ct) + \frac{1}{2} f(x-ct)\end{aligned}$$

where Euler's formula yields

$$\cos(\alpha ct) e^{i\alpha x} = \frac{e^{i\alpha ct} + e^{-i\alpha ct}}{2} e^{i\alpha x}.$$

Now, set $f(x) = 0$ (thus $F(\alpha) = 0$) to find the solution for zero initial displacement with an initial velocity $g(x)$:

$$\begin{aligned}
u(x, t) &= \int_{-\infty}^{\infty} G(\alpha) \frac{\sin(\alpha ct)}{c\alpha} e^{i\alpha x} d\alpha \\
&= \frac{1}{2c} \int_{-\infty}^{\infty} \frac{G(\alpha)}{i\alpha} e^{i\alpha(x+ct)} - \frac{G(\alpha)}{i\alpha} e^{i\alpha(x-ct)} d\alpha \\
&= \frac{1}{2c} \int_{-\infty}^{x+ct} g(s) ds - \frac{1}{2c} \int_{-\infty}^{x-ct} g(s) ds \\
&= \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds
\end{aligned}$$

where the Fourier transform's integration-division property yields the integrals in the spatial domain.

Use the principle of superposition to combine the displacement and velocity solutions to find the general solution:

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

This is known as **d'Alembert's solution**.

5.4 Convolution Theorem

The convolution theorem states that multiplication in the spatial domain is equivalent to convolution in the frequency domain:

$$\mathcal{F}[f(x)g(x)] = F(\alpha) * G(\alpha)$$

where the convolution is defined as

$$f(x) * g(x) \equiv \int_{-\infty}^{\infty} f(\tau)g(x - \tau) d\tau.$$

Note: this is assuming the definition of the Fourier transform presented in this reference. Since the placement of the constant $1/2\pi$ in the definition can vary, other versions may introduce an additional factor in the convolution theorem.

The convolution theorem also goes the other way around: multiplication in the frequency domain is equivalent to convolution in the spatial domain:

$$\mathcal{F}[f(x) * g(x)] = 2\pi F(\alpha)G(\alpha)$$

Notice that the 2π factor appears in this second equation due to the definition of the Fourier transform used in this reference.

6 Miscellaneous

6.1 Function Parity

A function $f(x)$ is **even** if $f(-x) = f(x)$ and **odd** if $f(-x) = -f(x)$.

- The product of two even functions is **even**.
- The product of two odd functions is **even**.
- The product of an even and an odd function is **odd**.

6.2 Dirac Delta Function

6.2.1 Definition

Given a rectangular pulse

$$d_\epsilon(x) = \begin{cases} \frac{1}{2\epsilon} & |x| \leq \epsilon \\ 0 & |x| > \epsilon \end{cases},$$

the Dirac delta function is defined as

$$\delta(x) = \lim_{\epsilon \rightarrow 0} d_\epsilon(x).$$

It is 0 everywhere in the domain except at $x = 0$ where it is infinitely tall. Alternatively, the Dirac delta function can be defined as

$$\delta(x) = \lim_{k \rightarrow \infty} d_k(x) = \lim_{k \rightarrow \infty} \frac{1}{\pi} \frac{k}{1 + x^2 k^2}$$

where $d_k(x)$ is the Lorentzian function (or Cauchy distribution) whose area under the curve is always 1 regardless of k .

6.2.2 Properties

By default, the impulse is at the origin, but it can be shifted:

$$\begin{aligned} \delta(x - x_0) & \quad (1\text{-dimensional}) \\ \delta(\mathbf{r} - \mathbf{r}_0) = \delta(x - x_0)\delta(y - y_0)\delta(z - z_0) & \quad (3\text{-dimensional}) \end{aligned}$$

Normalization of the Dirac delta function:

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(x) dx &= 1 & (1\text{-dimensional}) \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\mathbf{r}) dV &= 1 & (3\text{-dimensional}) \end{aligned}$$

A notable property of the Dirac delta function is the **sifting property**:

$$\int_{-\infty}^{\infty} f(x) \delta(x-a) dx = f(a) \quad (1\text{-dimensional})$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\mathbf{r}) \delta(\mathbf{r}-\mathbf{r}_0) dV = f(\mathbf{r}_0) \quad (3\text{-dimensional})$$

6.2.3 Derivatives

The first derivative of the Dirac delta function is

$$\delta'(x) = - \lim_{k \rightarrow \infty} \frac{d}{dx} d_k(x).$$

The normalization is given by

$$\int_{-\infty}^{\infty} \delta'(x) dx = 0.$$

The sifting property of the first derivative is given by

$$\int_{-\infty}^{\infty} f(x) \delta'(x-a) dx = -f'(a),$$

which can be generalized to

$$\int_{-\infty}^{\infty} f(x) \delta^{(n)}(x-a) dx = (-1)^n f^{(n)}(a).$$

6.3 Inner Product of Functions

The inner product of two functions $f(x)$ and $g(x)$ over the interval $[0, L]$ is given by

$$\langle f, g \rangle = \int_0^L f(x) g(x) dx.$$

If the functions are **orthogonal**, then their inner product is 0. Function orthogonality depends on the interval of integration.

The norm of a function is given by

$$\|f\| = \sqrt{\langle f, f \rangle} = \sqrt{\int_0^L f(x)^2 dx}$$

A function can be normalized by dividing it by its norm:

$$\hat{f}(x) = \frac{f(x)}{\|f\|} \implies \|\hat{f}\| = 1$$

A special case is with trigonometric functions:

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \begin{cases} \frac{L}{2} & n = m \\ 0 & n \neq m \end{cases}$$
$$\int_0^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx = \begin{cases} \frac{L}{2} & n = m \\ 0 & n \neq m \end{cases}$$